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|-----------------------------------------------------------------------------------|--------------|----------------------------------|--------------|-----------|
|  | Course Name: | Complex Variables and Transforms | Course Code: | MT2003    |
|                                                                                   | Program:     | BEE                              | Semester:    | Fall 2023 |
|                                                                                   | Duration:    | 3 hours                          | Total Marks: | 80        |
|                                                                                   | Exam Date:   | 27 <sup>th</sup> December, 2023  | Weight:      | 50        |
|                                                                                   | Section:     | ALL                              | Page(s):     | 2         |
|                                                                                   | Exam Type:   | Final Exam                       |              |           |

Student Name: \_\_\_\_\_ Roll No: \_\_\_\_\_ Section: \_\_\_\_\_

Instruction/Notes: 1. All CLOs are covered in this paper.  
2. If need arises, make valid assumptions and clearly mention it with your answer.

### Question 1 [10 marks, CLO1]

Electrical impedance is the ratio of voltage to current in ac circuits. Let  $Z$  represent the total impedance of an electrical circuit. If there are two resistors in a circuit are placed in parallel, the total impedance is given by following formula

$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

where  $Z_1 = 3 + 6i$  ohms and  $Z_2 = 5 + 4i$  ohms.

- [2 marks] Find the total impedance of the electrical circuit in terms of polar form.
- [5 marks] Use part (a) to find the fourth root of a given complex number and graph all roots in the complex plane. Show complete working.
- [3 marks] Use part (a) to find the principal value of  $\ln Z$ .

### Question 2 [10 marks, CLO2]

Verify that given function is harmonic? If yes then develop harmonic conjugate function and also write the corresponding analytic function in form of  $f(z) = u + iv$ .

$$u(x, y) = e^x \cos y + e^y \cos x + xy.$$

### Question 3 [30 marks, CLO3]

- [10 marks] Evaluate the given integral by using Cauchy integral formula along the given contour  $C$ , where  $C$  represents the circle  $C: |z|=2$  in counterclockwise direction.

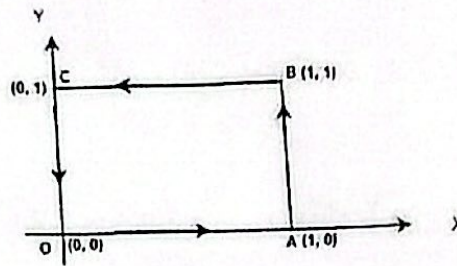
$$\oint_C \frac{ze^z}{z^2 + 1} dz$$

- [10 marks] Evaluate the given real integral by using the Cauchy Residue theorem and also mention the type.

$$\int_0^\pi \frac{1 + \cos 2\theta}{(5 + 4\cos\theta)} d\theta$$



- c. [10marks] Evaluate  $\oint_C |Z|^2 dz$  by using second evaluation method along the given contour C as shown in figure, where C consists of the boundaries of the squares counter clockwise. Show the complete working.



**Question 4 [30 marks, CLO4]**

- a. [10 marks] Find the Fourier series coefficients of the periodic function  $E(x)$ . Express  $E(x)$  as a Fourier series form.

$$E(x) = \begin{cases} \pi + x, & -\pi < x < 0 \\ \pi - x & 0 < x < \pi \end{cases}$$

- b. [10 marks] Find Fourier cosine integral representation of the given function

$$f(x) = \begin{cases} x^2, & 0 < x < a \\ 0 & x > a \end{cases}$$

- c. [10 marks] Find Fourier sine transforms of the signals represented by the given function

$$f(x) = \begin{cases} 3, & 0 < x < 2 \\ 2 - x, & 2 < x < 4 \\ 0 & x > 4 \end{cases}$$